

Jaypee Institute of Information Technology, Noida

Department of Computer Science & Engineering and IT



Major Project Title: Human Body Mass Index Prediction from Images
Using Pose Estimation and Machine Learning

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8th Sem

2025 - 2026

DECLARATION

I **Jaspal Singh (22103191)** hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

Signature of Student(s): _____

Date: April 25, 2026

CERTIFICATE

This is to certify that the work titled **Human Body Mass Index Prediction from Images Using Pose Estimation and Machine Learning** was submitted by **Jaspal Singh** in partial fulfilment for the award of the degree of B.Tech of Jaypee Institute of Information Technology, Noida has been carried out under my supervision. This work has not been submitted partially or wholly to any other University or Institute for the award of this or any other degree or diploma.

Signature of the Supervisor:_____

Name of the Supervisor: **Dr. Anil Kumar Mahto**

Date: April 25 , 2026

ACKNOWLEDGEMENT

We would like to place on record our deep sense of gratitude to **Dr. Anil Kumar Mahto**, Jaypee Institute of Information Technology, Noida for his generous guidance, help and useful suggestions. We express our sincere gratitude to him for his stimulating guidance, continuous encouragement and supervision throughout the course of the present work

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SUMMARY

This project presents an **AI-based Body Mass Index (BMI) estimation system** that predicts a person's BMI using body posture images instead of traditional height and weight measurements. The system combines computer vision and machine learning techniques to extract meaningful body features and estimate BMI in a non-contact, automated manner.

The process begins with **pose estimation using MediaPipe**, where key body landmarks such as shoulders, hips, arms, and legs are detected from an input image or live camera feed. Based on these landmarks, several geometric features are calculated, including body proportions and ratios like shoulder-to-hip ratio and height-normalized measurements. These features represent the overall body structure and are used as input to the machine learning model.

A **Random Forest Regression model** is trained on a synthetically generated dataset that simulates different body types across BMI ranges such as underweight, normal, overweight, and obese. The dataset is carefully balanced to ensure that the model learns patterns from all categories instead of being biased toward a specific body type. During training, the model learns how variations in body proportions relate to BMI values.

The system includes a **Flask-based web application** that allows users to upload images or use a live camera for real-time BMI prediction. The interface displays the predicted BMI value, corresponding health category, and a confidence score. Additionally, a history feature stores previous predictions, enabling users to track changes over time.

To improve reliability, the project incorporates feature scaling, prediction smoothing, and rule-based corrections that align model outputs with realistic human body characteristics. The system also visualizes **feature importance**, helping to understand which body measurements contribute most to BMI prediction.

Overall, this project demonstrates how AI can be used to create a practical, accessible, and non-invasive health assessment tool. It highlights the integration of computer vision, machine learning, and web technologies to solve a real-world problem while providing a foundation for future improvements such as multi-model comparison, real-world dataset integration, and enhanced accuracy.

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CHAPTER 1

INTRODUCTION

1.1 General Introduction

Body Mass Index (BMI) is a widely used indicator to assess whether a person has a healthy body weight relative to their height. Traditionally, BMI is calculated using physical measurements such as height and weight. However, obtaining these measurements requires physical contact, proper instruments, and sometimes clinical supervision, which may not always be convenient or accessible.

With the advancement of artificial intelligence and computer vision, it has become possible to estimate physical characteristics from images. This project focuses on developing an **AI-based BMI estimation system** that predicts BMI using body posture images. By utilizing pose detection techniques, key body landmarks such as shoulders, hips, and limbs are identified, and meaningful features are extracted.

These features are then used as input to machine learning models to estimate BMI. The system supports both image upload and live camera input, making it practical and user-friendly. This approach demonstrates how AI can provide a non-contact and automated solution for basic health assessment.

1.2 Problem Statement

The traditional method of calculating BMI requires accurate measurement of height and weight, which is not always feasible in remote or resource-limited environments. Additionally, manual measurement can be time-consuming and dependent on proper equipment.

The problem addressed in this project is:

- To develop a system that can estimate BMI accurately using only an image or live camera input, without requiring direct physical measurements.

The challenge lies in extracting reliable body features from images and mapping them effectively to BMI values using machine learning techniques.

1.3 Significance / Novelty of the Problem

The increasing prevalence of obesity and related health issues has made Body Mass Index (BMI) an important indicator for assessing an individual's health status. Traditionally, BMI is calculated using a person's weight and height, requiring physical measurements and, in some cases, specialized equipment. While this method is accurate, it is not always convenient, especially in situations where quick, non-invasive, or remote health assessment is required.

The significance of this problem lies in the need for an accessible, fast, and user-friendly method of estimating BMI without relying on manual measurements. In today's digital era, where smartphones and cameras are widely available, there is a growing opportunity to leverage computer vision and artificial intelligence to provide alternative solutions for health monitoring. This project addresses that need by proposing a system that estimates BMI directly from images or live video input.

The novelty of the proposed approach lies in the integration of pose estimation and machine learning techniques to infer BMI from body proportions. Using frameworks such as Google AI's MediaPipe, the system extracts key human body landmarks and derives meaningful features such as shoulder width, hip width, torso length, and limb proportions. These features are then processed using a machine learning model, specifically a Random Forest algorithm introduced by Leo Breiman, to predict BMI values.

Unlike traditional BMI calculators, which depend solely on numerical inputs, this system interprets visual information to estimate body composition. This makes the approach non-invasive and more intuitive, as users can simply provide an image or use a live camera feed. Additionally, the use of real-time processing enhances usability, allowing continuous monitoring and immediate feedback.

Another novel aspect of the project is the use of synthetic yet structured datasets that mimic real human body proportions. By carefully designing the dataset to reflect realistic variations in body shape and fat distribution, the system ensures that the machine learning model learns meaningful patterns rather than relying on arbitrary data. Feature engineering techniques, such as normalization and ratio-based measurements, further improve the robustness and interpretability of the model.

Furthermore, the integration of a web-based interface using Flask enables seamless interaction between the user and the system. This allows the solution to be easily deployed and accessed without requiring complex installations, thereby increasing its practical applicability.

In summary, the proposed system is significant because it provides a convenient and scalable alternative to traditional BMI measurement methods. Its novelty lies in combining computer vision, machine learning, and real-time processing to deliver a non-contact, AI-driven health assessment tool. This approach not only enhances accessibility but also demonstrates the potential of artificial intelligence in transforming conventional healthcare practices.

1.4 Empirical Study

The empirical study conducted for this project focuses on evaluating the feasibility and effectiveness of estimating Body Mass Index (BMI) using computer vision and machine learning techniques. The study involves collecting data, designing a dataset, training a predictive model, and validating its performance through experimental analysis.

To begin with, a dataset was created to simulate realistic human body proportions corresponding to different BMI categories such as underweight, normal, overweight, and obese. Since obtaining a large-scale real-world dataset with annotated BMI values is challenging, a synthetic dataset was generated based on known relationships between body structure and BMI. Key features such as shoulder width, hip width, torso length, arm length, and leg length were derived, along with normalized ratios like shoulder-to-height and hip-to-height. Care was taken to introduce controlled randomness in the dataset to reduce excessive correlation between features and ensure that the model learns meaningful patterns.

The feature extraction process in the empirical study utilizes pose estimation techniques provided by Google AI's MediaPipe framework. This allows the system to detect human body landmarks from images or live video input and compute the same set of features used during training. Ensuring consistency between training data and real-time input is a critical aspect of the study.

For prediction, a Random Forest regression model, introduced by Leo Breiman, was employed due to its robustness and ability to handle nonlinear relationships between features. The dataset was divided into training and testing sets to evaluate the model's performance objectively. Performance metrics such as Mean Absolute Error (MAE) and R^2 score were used to measure prediction accuracy and reliability.

Experimental results indicate that the model is capable of estimating BMI with a high degree of accuracy under controlled conditions. The feature importance analysis shows that hip width plays a dominant role in BMI prediction, followed by torso and other body measurements. This aligns with

real-world observations, where fat distribution around the hips and abdomen significantly influences BMI.

The empirical study also includes testing the system using real images and live camera input. Multiple test cases were analyzed to observe how variations in posture, clothing, and camera angle affect prediction accuracy. It was observed that proper body visibility and alignment improve results, while occlusions or improper framing can lead to minor deviations.

Overall, the empirical study demonstrates that BMI can be reasonably estimated using visual data and machine learning. While the system may not replace traditional measurement methods in medical settings, it provides a practical and efficient alternative for general health monitoring. The findings validate the effectiveness of the proposed approach and highlight its potential for further improvement and real-world application

1.5 Brief Description of the Solution Approach

The proposed system follows a step-by-step pipeline:

1.5.1. Input Acquisition

- User uploads an image or uses a live camera

1.5.2. Pose Detection

- MediaPipe identifies body landmarks such as shoulders, hips, arms, and legs

1.5.3. Feature Extraction

- Measurements and ratios are calculated from detected landmarks

1.5.4. Machine Learning Prediction

- Extracted features are passed to trained models (e.g., Random Forest)

1.5.5. Result Display

- BMI value, category (Underweight, Normal, Overweight, Obese), and confidence score are shown

1.5.6. History Tracking

- Previous predictions are stored for analysis

This approach integrates computer vision and machine learning into a single system.

1.6 Comparison of Existing Approaches

Traditional BMI calculation methods rely on direct measurement of height and weight, which require physical interaction and equipment. Some advanced approaches use wearable devices or medical imaging, which can be costly and less accessible.

- In contrast, the proposed system:
- Requires only an image or camera input
- Does not depend on physical measurement tools
- Provides quick and automated results
- Is cost-effective and easy to deploy

While existing methods may offer higher accuracy, the proposed solution provides a **practical and accessible alternative** for general health assessment. It balances convenience and accuracy by leveraging AI techniques.

CHAPTER 2

LITERATURE SURVEY

2.1 Summary of Papers Studied

A number of research works related to computer vision, pose estimation, and machine learning-based prediction systems were studied to understand the feasibility of estimating Body Mass Index (BMI) using image data. The following section summarizes the key contributions of relevant papers and how they relate to the proposed system

Paper 2.1: MediaPipe: A Framework for Building Perception Pipelines

Authors: Camillo Lugaresi et al.

Year: 2019

Publisher: Google Research

This paper introduces MediaPipe, a framework designed for building real-time perception pipelines. It supports various tasks such as face detection, hand tracking, and pose estimation. The framework is optimized for performance and can run efficiently on devices with limited computational resources.

Use in Project:

MediaPipe is used in this project to detect human body landmarks. It forms the core component for extracting body features required for BMI estimation

Paper 2.2: BlazePose: On-device Real-time Body Pose Tracking

Authors: Valentin Bazarevsky et al.

Year: 2020

Publisher: Google AI

BlazePose is a lightweight deep learning model capable of detecting 33 human body landmarks in real time. It is designed for high accuracy and efficiency, especially for mobile and embedded applications.

Use in Project:

The pose detection in this system is based on BlazePose through MediaPipe, enabling accurate extraction of body measurements such as shoulder width and hip width.

Paper 2.3: Visual Estimation of BMI from Facial Images

Authors: Wen and Guo

Year: 2013

Publisher: IEEE

This research explores estimating BMI using facial features extracted from images. It demonstrates that certain visual characteristics can be correlated with BMI values using machine learning techniques.

Use in Project:

This study provides the foundational idea that BMI can be estimated visually. The current project extends this concept by using full-body pose features instead of only facial features.

Paper 2.4: Predicting BMI from Body Shape Using Machine Learning

Authors: Pfitzner et al.

Year: 2018

Publisher: Springer

The paper focuses on predicting BMI using body shape parameters and regression models. It highlights the importance of geometric body features in determining BMI.

Use in Project:

This work supports the use of body proportions such as torso length and hip width, which are also used as key features in this project.

Paper 2.5: Random Forests

Authors: Leo Breiman

Year: 2001

Publisher: Machine Learning Journal

This paper introduces the Random Forest algorithm, an ensemble learning technique that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Use in Project:

Random Forest is used as the primary machine learning model for BMI prediction due to its ability to handle nonlinear relationships and feature interactions effectively.

Paper 2.6: Support Vector Machines for Regression

Authors: Vladimir Vapnik et al.

Year: 1995

Publisher: Springer

This work explains Support Vector Regression (SVR), which is effective in modeling complex relationships in data using kernel functions.

Use in Project:

SVR is implemented as a comparative model to evaluate performance against Random Forest and Linear Regression.

Paper 2.7: Multiple Linear Regression Analysis

Authors: Draper and Smith

Year: 1998

Publisher: Wiley

This book explains linear regression techniques used to model relationships between dependent and independent variables.

Use in Project:

Linear Regression is used as a baseline model to compare the performance of more advanced machine learning models.

Paper 2.8: Human Pose Estimation: A Survey

Authors: Sarafianos et al.

Year: 2016

Publisher: IEEE

This survey reviews various approaches to human pose estimation, including both traditional and deep learning methods. It discusses challenges such as accuracy, speed, and robustness.

Use in Project:

Provides background understanding for selecting pose estimation as the primary technique for extracting body features.

Paper 2.9: Deep Learning for Computer Vision: A Review

Authors: Rawat and Wang

Year: 2017

Publisher: Neural Computation

This paper reviews deep learning techniques used in computer vision tasks such as object detection and image classification.

Use in Project:

Supports the use of computer vision techniques for extracting meaningful information from images in healthcare-related applications.

Paper 2.10: Applications of Machine Learning in Healthcare

Authors: Jiang et al.

Year: 2017

Publisher: IEEE

This paper discusses the role of machine learning in healthcare, including prediction systems and decision support tools.

Use in Project:

The proposed BMI estimation system is an application of machine learning in healthcare for basic health monitoring.

2.2 Integrated Summary of the Literature Studied

The literature reviewed for this project spans three major domains: computer vision and pose estimation, machine learning techniques for prediction, and the use of Body Mass Index (BMI) in healthcare applications. Integrating insights from these areas provides a strong foundation for the development of an AI-based BMI estimation system.

Studies on pose estimation techniques, particularly those involving frameworks like Google AI's MediaPipe and OpenPose, demonstrate that human body landmarks can be accurately detected in real time using RGB images. These methods enable the extraction of key body measurements such as shoulder width, hip width, and limb lengths without requiring specialized hardware. This capability forms the backbone of the proposed system, where visual input is converted into meaningful numerical features.

From a machine learning perspective, algorithms such as Random Forests, introduced by Leo Breiman, have been widely used for regression and prediction tasks due to their robustness and ability to handle nonlinear relationships. Literature suggests that ensemble models perform well when dealing with structured data and engineered features, making them suitable for BMI prediction based on body proportions. Additionally, tools like Scikit-learn provide efficient implementations that simplify model development and evaluation.

In the healthcare domain, BMI remains a widely accepted indicator for categorizing individuals into underweight, normal, overweight, and obese groups, as defined by organizations such as the World Health Organization and the Centers for Disease Control and Prevention. However, traditional BMI calculation requires physical measurements of height and weight, which may not always be convenient or accessible. Recent research highlights the potential of AI-driven, non-invasive approaches to estimate BMI using visual data, thereby improving accessibility and user experience.

By combining these insights, the proposed system leverages pose estimation to extract anthropometric features and applies machine learning to predict BMI. Unlike traditional methods, this approach eliminates the need for manual input and enables real-time analysis using images or live video. Furthermore, the integration of feature engineering techniques, such as normalization and ratio computation, enhances the model's ability to capture meaningful relationships between body structure and BMI.

Overall, the literature indicates a clear trend toward the use of artificial intelligence in healthcare, particularly for non-invasive monitoring and analysis. The proposed project aligns with this trend by

offering a practical, scalable, and user-friendly solution for BMI estimation. It bridges the gap between theoretical research and real-world application by combining established techniques from multiple domains into a unified system.

CHAPTER 3

REQUIREMENT ANALYSIS AND SOLUTION APPROACH

3.1 Overall Description of the Project

The **AI-Based BMI Estimation System** is a computer vision and machine learning based application developed to estimate a person's Body Mass Index using full-body images or live camera input. Traditionally, BMI is calculated using height and weight values entered manually by the user. However, this project attempts to automate the process by analyzing body proportions from an image and predicting BMI using a trained machine learning model.

The system uses **MediaPipe Pose Estimation** to detect important body landmarks such as shoulders, hips, arms, legs, and feet. These landmarks are processed to extract numerical body features such as shoulder width, hip width, body height, torso length, arm length, leg length, and body proportion ratios. These extracted features are then passed to a trained Random Forest Regression model, which predicts the BMI value.

The application is built using **Python Flask** for the backend and web interface. It also includes a live camera module using **OpenCV**, allowing real-time BMI estimation through webcam input. The predicted BMI is classified into standard categories such as **Underweight, Normal, Overweight, and Obese**. The system also stores previous BMI predictions and displays BMI trends using charts.

System Components

Frontend Interface

The frontend provides a simple web interface where users can upload images and view BMI results. It includes:

- Image upload option
- Image preview
- BMI prediction button
- Live camera button
- BMI result display
- BMI category display
- Reliability score
- BMI history table
- BMI trend chart

The interface is developed using **HTML, CSS, JavaScript, Jinja templates**, and **Chart.js**.

Backend Processing

The backend is developed using **Flask**. It handles image upload, file validation, feature extraction, model prediction, history storage, and result rendering.

Main backend responsibilities include:

- Handling user requests
- Saving uploaded images
- Calling pose estimation module
- Calling trained BMI prediction model
- Categorizing BMI result
- Saving prediction history
- Displaying results on the web page

Computer Vision Processing

The computer vision part uses **OpenCV** and **MediaPipe**. OpenCV is used to read images and access the webcam, while MediaPipe is used to detect human body landmarks and segmentation masks.

Machine Learning Model

The machine learning component uses a **Random Forest Regressor** trained on body proportion features. The trained model is saved as ``model.pkl`` and loaded during prediction.

Data Storage

The project uses local file-based storage:

- ``dataset.csv`` stores training data
- ``history.csv`` stores BMI prediction history
- ``model.pkl`` stores the trained model
- ``static/uploads/`` stores uploaded images

Product Functions

The AI-Based BMI Estimation System provides the following core features:

Image-Based BMI Estimation

The user uploads a full-body image, and the system predicts BMI using body measurements extracted from the image.

Live Camera BMI Detection

The system uses webcam input to detect body landmarks in real time and estimate BMI.

Pose Detection

MediaPipe detects human body landmarks required for body proportion analysis.

Body Feature Extraction

The system extracts features such as shoulder width, hip width, height, torso, arm length, leg length, and body ratios.

BMI Prediction

The trained Random Forest model predicts BMI from extracted features.

BMI Classification

The predicted BMI is categorized as:

- Underweight
- Normal
- Overweight
- Obese

Reliability Score

The system calculates a reliability score based on whether extracted body features are within expected measurement ranges.

BMI History and Trend

The system stores previous BMI predictions and displays them in tabular and graphical form.

3.2 Requirement Analysis

3.2.1 Functional Requirements

FR1 – Image Upload

- FR1.1: The system shall allow users to upload full-body images.
- FR1.2: The system shall support JPG, JPEG, and PNG image formats.
- FR1.3: The system shall reject unsupported file types.
- FR1.4: The uploaded image shall be saved in the upload folder.
- FR1.5: The uploaded image shall be displayed as a preview.

FR2 – Pose Detection

- FR2.1: The system shall detect human body landmarks from the uploaded image.
- FR2.2: The system shall check whether the full body is visible.
- FR2.3: The system shall detect shoulders, hips, arms, legs, and feet.
- FR2.4: The system shall display an error if no pose is detected.

FR3 – Feature Extraction

- FR3.1: The system shall extract body measurement features from pose landmarks.
- FR3.2: The system shall calculate shoulder width.
- FR3.3: The system shall calculate hip or body width.
- FR3.4: The system shall calculate normalized body height.
- FR3.5: The system shall calculate torso, arm, and leg measurements.
- FR3.6: The system shall calculate shoulder-to-hip and hip-to-height ratios.

- FR3.7: The extracted feature order shall match the training feature order.

FR4 – BMI Prediction

- FR4.1: The system shall load the trained model from `model.pkl`.
- FR4.2: The system shall pass extracted features to the model.
- FR4.3: The system shall predict BMI as a numerical value.
- FR4.4: The system shall restrict BMI values within a reasonable range.
- FR4.5: The final BMI value shall be rounded before display.

FR5 – BMI Classification

- FR5.1: BMI below 18.5 shall be classified as Underweight.
- FR5.2: BMI from 18.5 to 24.9 shall be classified as Normal.
- FR5.3: BMI from 25.0 to 29.9 shall be classified as Overweight.
- FR5.4: BMI 30.0 and above shall be classified as Obese.

FR6 – Live Camera Detection

- FR6.1: The system shall allow users to start live camera mode.
- FR6.2: The system shall capture webcam frames using OpenCV.
- FR6.3: The system shall detect pose landmarks in live frames.
- FR6.4: The system shall estimate BMI from live camera input.
- FR6.5: The system shall display BMI, category, and reliability on the camera window.
- FR6.6: The system shall reject poor camera frames where the full body is not visible.

FR7 – Reliability Score

- FR7.1: The system shall calculate reliability based on extracted feature quality.
- FR7.2: The system shall check whether features are within expected ranges.
- FR7.3: The system shall reduce reliability for abnormal measurements.
- FR7.4: The reliability score shall be shown with the BMI result.

FR8 – BMI History

- FR8.1: The system shall store predicted BMI values.
- FR8.2: The system shall store the BMI category.
- FR8.3: The system shall display previous BMI predictions.
- FR8.4: The system shall allow users to clear BMI history.

FR9 – Model Training

- FR9.1: The system shall load training data from `dataset.csv`.
- FR9.2: The system shall train a Random Forest Regression model.
- FR9.3: The dataset shall be split into training and testing data.
- FR9.4: The model shall be evaluated using MAE and R2 score.
- FR9.5: The trained model shall be saved as `model.pkl`.

3.2.2 Non-Functional Requirements

NFR1 – Performance

- NFR1.1: Image-based BMI prediction should complete within a few seconds.
- NFR1.2: The web page should load quickly.
- NFR1.3: Live camera mode should process frames smoothly.
- NFR1.4: Model prediction should be lightweight for local execution.

NFR2 – Accuracy

- NFR2.1: The system should provide reasonable BMI estimates.
- NFR2.2: Prediction should not be performed when pose detection fails.
- NFR2.3: Body segmentation should improve body width estimation.
- NFR2.4: Model accuracy should be evaluated using proper metrics.

NFR3 – Reliability

- NFR3.1: The system should handle missing images without crashing.
- NFR3.2: The system should handle failed pose detection gracefully.
- NFR3.3: Invalid feature vectors should not be sent to the model.
- NFR3.4: The system should display clear error messages.

NFR4 – Usability

- NFR4.1: The interface should be simple and easy to understand.
- NFR4.2: Users should be able to upload images easily.
- NFR4.3: BMI categories should be displayed using clear colors.
- NFR4.4: BMI history and trend should be easy to read.

NFR5 – Maintainability

- NFR5.1: The code should be modular.
- NFR5.2: Training, prediction, feature extraction, and camera logic should be separated.
- NFR5.3: Functions should be reusable.
- NFR5.4: Project dependencies should be listed in `requirements.txt`.

NFR6 – Security

- NFR6.1: Only valid image files should be accepted.
- NFR6.2: Uploaded filenames should be sanitized.
- NFR6.3: Uploaded files should not be executed.
- NFR6.4: Images should be stored only in the defined upload folder.

3.2.3 Logical Data Storage Requirements

Dataset File

`dataset.csv` stores training data.

```

```json
{
 "shoulder": "Float",
 "hip": "Float",
 "height": "Float",
 "torso": "Float",
 "arm": "Float",
 "leg": "Float",
 "ratio": "Float",
 "shoulder_h": "Float",
 "hip_h": "Float",
 "bmi": "Float"
}
```

```

History File

`history.csv` stores previous BMI predictions.

```

```json
{
 "BMI": "Predicted BMI value",
 "Category": "BMI category"
}
```

```

Model File

`model.pkl` stores the trained Random Forest model.

```

```json
{
 "model": "Trained RandomForestRegressor",
 "purpose": "BMI prediction"
}
```

```

Upload Folder

`static/uploads/` stores uploaded images.

```

```json
{
 "image": "Uploaded full-body image",
 "purpose": "Pose detection and BMI prediction"
}
```

```

Data Integrity Rules

The feature order must remain same during training and prediction.

- `dataset.csv` must contain all required feature columns.

- `history.csv` must contain BMI and Category fields.
- Uploaded files must be valid image files.
- Prediction must not run if pose detection fails.
- Feature vector length must be exactly 9.
- The saved model must be compatible with the current scikit-learn version.

3.3 Solution Approach

3.3.1 Overall Architecture

The solution follows a modular architecture with four major layers:

1. Presentation Layer
2. Application Layer
3. Computer Vision and Feature Extraction Layer
4. Machine Learning Layer

1. Presentation Layer

The presentation layer is the web interface of the system. It allows users to upload images, view results, use live camera mode, and check BMI history.

It includes:

- Upload form
- Image preview
- Result card
- BMI category display
- Reliability score
- BMI history table
- BMI trend graph

2. Application Layer

The application layer is handled by Flask. It manages routes, file uploads, feature extraction calls, prediction calls, and result rendering.

Main file:

```
```text
app.py
```
```

3. Computer Vision and Feature Extraction Layer

This layer detects body landmarks and calculates body features.

Main files:

```
```text
pose_estimation.py
body_features.py
live_camera.py
```
```

It uses MediaPipe and OpenCV for pose estimation and body feature extraction.

4. Machine Learning Layer

This layer predicts BMI using the trained Random Forest model.

Main files:

```
```text
train_model.py
evaluate_model.py
bmi_model.py
model.pkl
```
```

3.3.2 Module-Wise Detailed Description

Module 1: Web Application Module

Purpose:

To provide a user interface for uploading images and viewing BMI results.

Workflow:

- User opens the Flask web application.
- User uploads a full-body image.
- System validates and saves the image.
- System extracts body features.
- Model predicts BMI.
- Result is displayed on the web page.

Module 2: Pose Estimation Module

Purpose:

To detect human body landmarks from an image.

Main file:

```
```text
pose_estimation.py
```
```

Workflow:

Image is read using OpenCV.
Image is converted to RGB.
MediaPipe Pose detects landmarks.
If landmarks are missing, an error is returned.
Landmarks are passed to the feature extraction module.

Module 3: Body Feature Extraction Module

Purpose:
To convert pose landmarks into numerical features.

Main file:

```
```text  
body_features.py
```
```

Features extracted:

- Shoulder width
- Hip width
- Height
- Torso length
- Arm length
- Leg length
- Shoulder-to-hip ratio
- Shoulder-to-height ratio
- Hip-to-height ratio

Module 4: BMI Prediction Module

Purpose:
To predict BMI using the trained model.

Main file:

```
```text  
bmi_model.py
```
```

Workflow:

- Load `model.pkl`.
- Validate feature length.
- Convert features into DataFrame.
- Predict BMI.
- Return final BMI value.

Module 5: Dataset Generation Module

Purpose:
To generate training data for the BMI model.

Main file:

```
```text
generate_dataset.py
```
```

The dataset contains generated body proportion features and BMI labels for different BMI categories.

Module 6: Model Training Module

Purpose:

To train the Random Forest Regression model.

Main file:

```
```text
train_model.py
```
```

Workflow:

- Load dataset
- Select features
- Split train/test data
- Train Random Forest model
- Evaluate performance
- Save model

Module 7: Model Evaluation Module

Purpose:

To evaluate trained model performance.

Main file:

```
```text
evaluate_model.py
```
```

Metrics used:

- Mean Absolute Error
- R2 Score

Generated files:

```
```text
evaluation_plot.png
feature_importance.png
```
```

Module 8: Live Camera Module

Purpose:

To estimate BMI using webcam input.

Main file:

```
```text
live_camera.py
```
```

Workflow:

- Open webcam
- Capture frames
- Detect pose
- Extract features
- Predict BMI
- Display result on camera window

Module 9: History and Reporting Module

Purpose:

To store and display previous BMI predictions.

Files used:

```
```text
history.csv
templates/index.html
```
```

Functions:

- Save BMI result
- Display previous results
- Show BMI trend chart
- Clear history

3.3.3 Technology Stack

Frontend Technologies

- HTML: Used for page structure.
- CSS: Used for styling the interface.
- JavaScript: Used for image preview.
- Jinja Templates: Used for dynamic result rendering.
- Chart.js: Used for BMI trend visualization.

Backend Technologies

- Python: Main programming language.
- Flask: Web framework for routing and backend logic.
- Werkzeug: Used for secure file upload handling.

Computer Vision Technologies

- OpenCV: Used for image reading, webcam access, and frame display.
- MediaPipe Pose: Used for body landmark detection.
- MediaPipe Segmentation: Used for body outline estimation.

Machine Learning Technologies

- Scikit-learn: Used for model training and evaluation.
- Random Forest Regressor: Used for BMI prediction.
- Pandas: Used for CSV data handling.
- NumPy: Used for numerical operations.
- Pickle: Used for saving and loading the trained model.

Data Storage

- CSV files: Used for dataset and history storage.
- Local upload folder: Used for storing uploaded images.
- Pickle file: Used for storing trained model.

Development Tools

- Visual Studio Code: Used for coding.
- Python Virtual Environment: Used for dependency management.
- Browser: Used for testing Flask web application.
- Git/GitHub: Can be used for version control.

3.3.4 Algorithm Used

The main algorithm used in this project is Random Forest Regression.

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees. Each tree makes its own prediction, and the final result is calculated by averaging the predictions of all trees.

Reasons for Choosing Random Forest

- Handles nonlinear relationships well
- Works effectively with tabular data
- Reduces overfitting compared to a single decision tree
- Provides feature importance values
- Does not require complex preprocessing
- Suitable for small and medium-sized datasets

Input Features

```
```text
shoulder
hip
height
torso
arm
leg
ratio
shoulder_h
hip_h
```
```

Output

```
```text
Predicted BMI value
```
```

3.3.5 BMI Classification Logic

| BMI Range | Category |
|----------------|-------------|
| Below 18.5 | Underweight |
| 18.5 to 24.9 | Normal |
| 25.0 to 29.9 | Overweight |
| 30.0 and above | Obese |

3.3.6 Proposed Workflow

1. User uploads a full-body image.
2. Flask validates the image.
3. Image is saved in upload folder.
4. OpenCV reads the image.
5. MediaPipe detects body landmarks.
6. Body features are extracted.
7. Features are passed to the trained model.
8. BMI is predicted.
9. BMI category is assigned.
10. Reliability score is calculated.
11. Result is shown on the web page.
12. BMI history is saved.
13. BMI trend chart is updated.

3.3.7 Limitations

- BMI prediction from image is only an approximation.
- Loose clothing may affect body width detection.
- Poor lighting can reduce accuracy.
- Side poses can produce incorrect results.
- Camera angle affects measurements.
- Muscular people may be classified as overweight because BMI does not distinguish fat from muscle.
- The system is not a replacement for medical diagnosis.
- Real-world accuracy depends on dataset quality.

3.3.8 Future Scope

- Use real labelled image datasets.
- Add manual height input for better accuracy.
- Improve body segmentation.
- Add user login system.
- Store results in a database.
- Generate downloadable BMI reports.
- Add mobile camera support.
- Train with more diverse body types.
- Improve live camera stability.
- Add recommendation system based on BMI category.

CHAPTER 4

MODELLING AND IMPLEMENTATION DETAILS

4.1 Design Diagrams

4.1.1 Use Case Diagrams

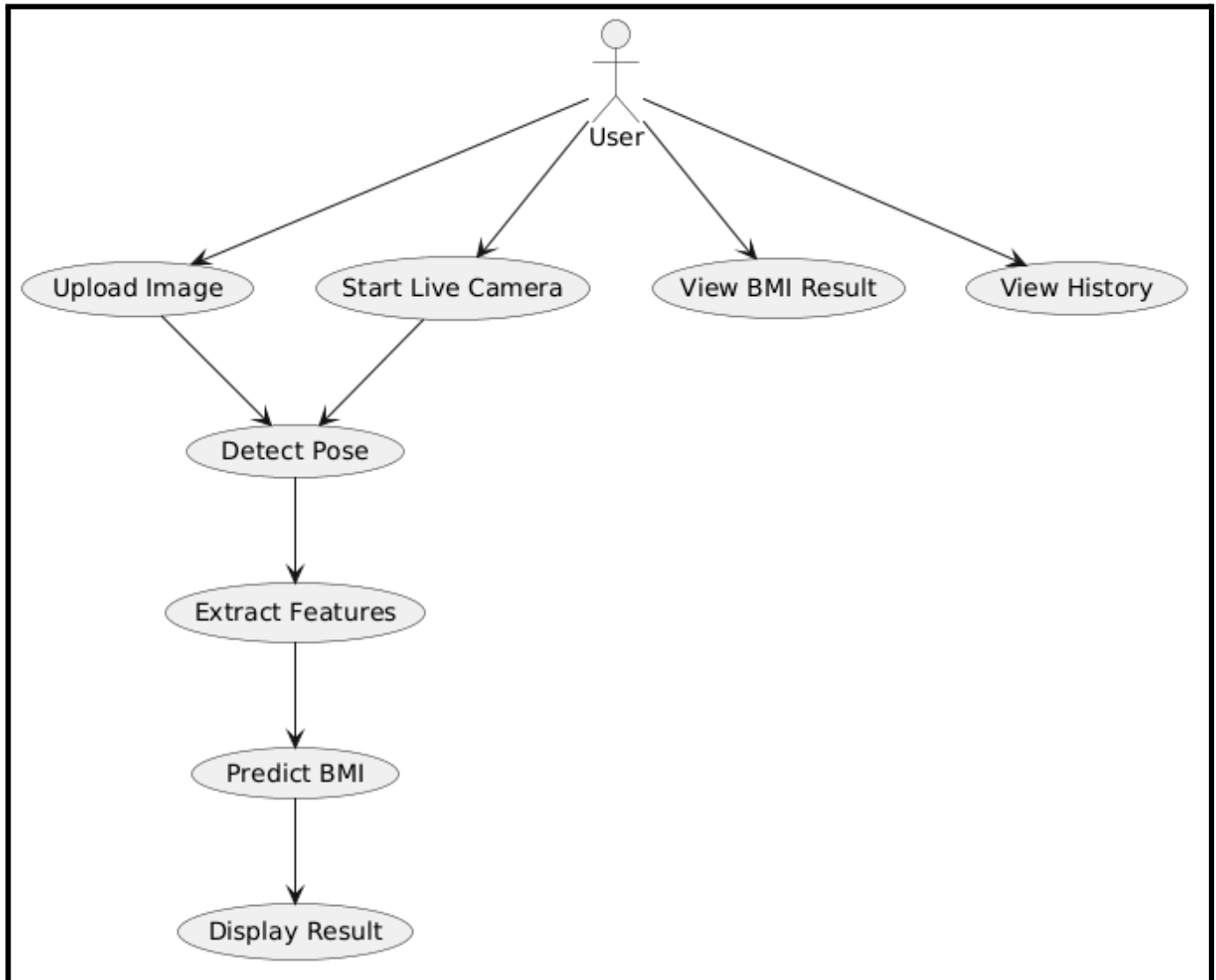


Fig 4.1 Use Case Diagram

4.1.2 Class Diagrams

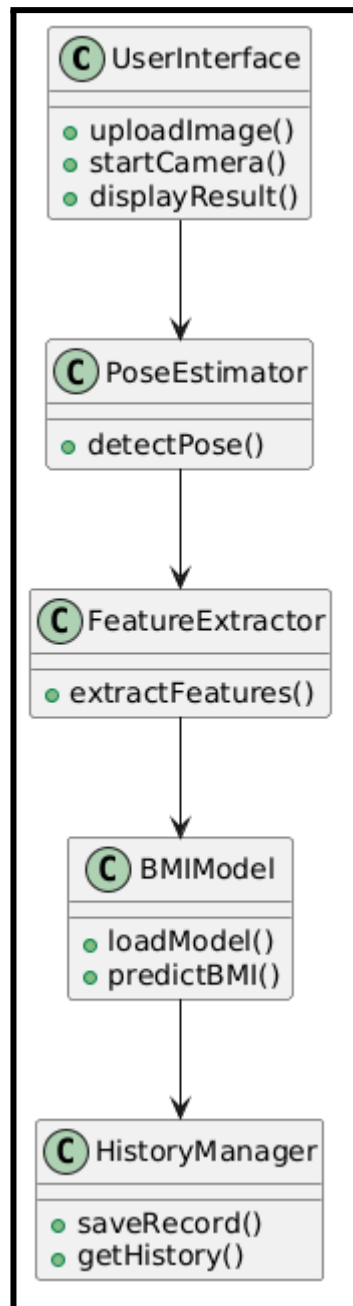


Fig 4.2 Class Diagram

4.1.3 Sequence Diagram

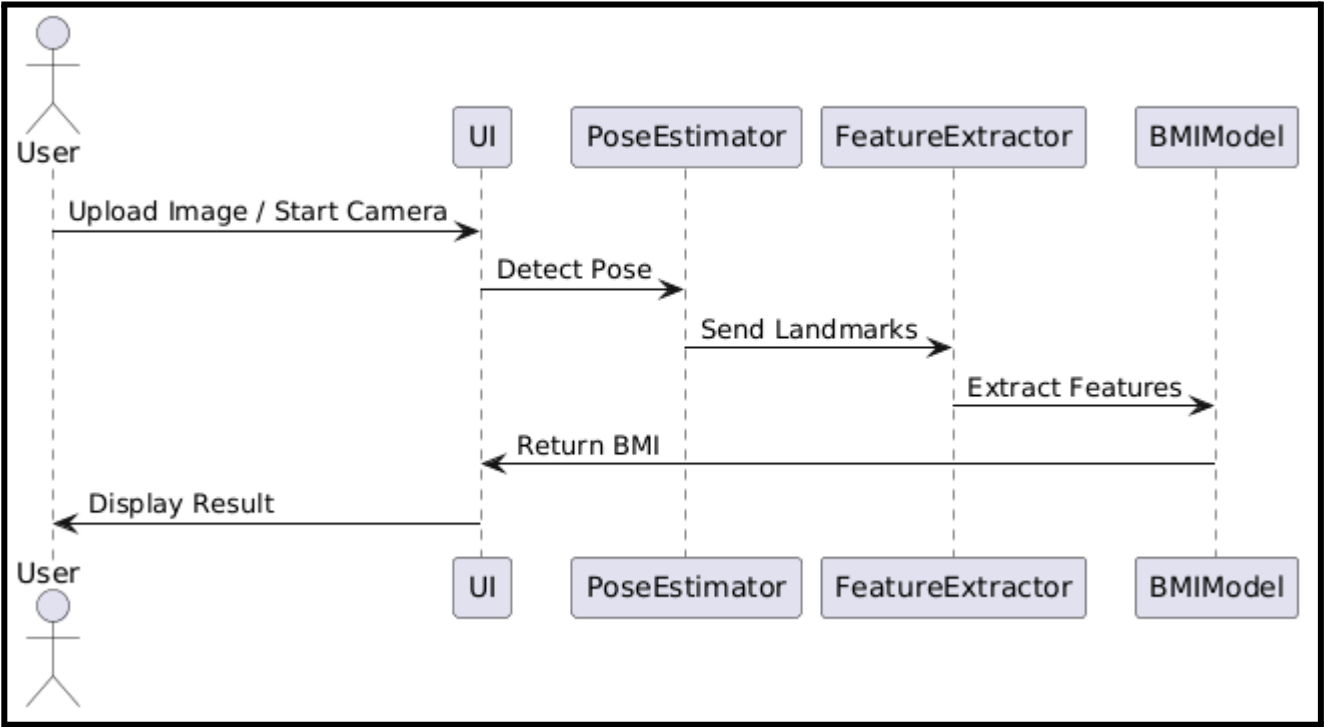


Fig 4.3: Sequence Diagram

4.1.4 System Flow Diagram

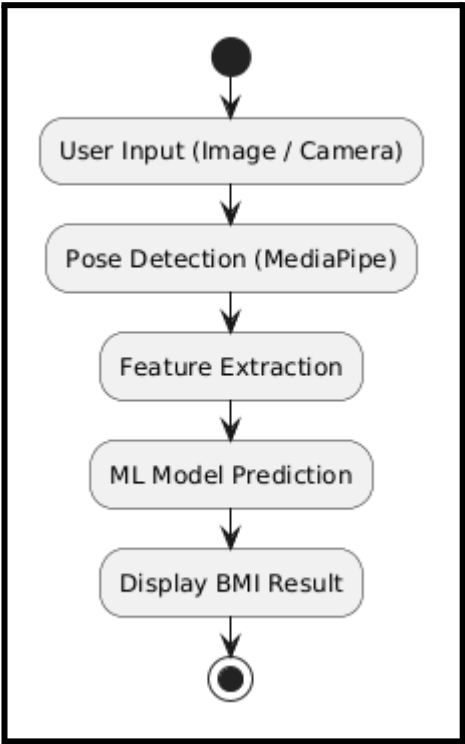


Fig. 4.4 System Flow Diagram

4.2 Implementation Details and Issues

Key Implementation Decisions

4.2.1. Use of MediaPipe for Pose Detection

Decision:

Use MediaPipe instead of building a custom pose detection model.

Rationale:

MediaPipe provides accurate and real-time pose detection without requiring heavy computational resources or large datasets. It simplifies development and ensures reliable landmark detection.

4.2.2. Use of Synthetic Dataset for Training

Decision:

Generate dataset instead of collecting real-world data.

Rationale:

Collecting real BMI-labeled image data is difficult and time-consuming. A synthetic dataset allows controlled feature generation and balanced representation of BMI categories.

4.2.3. Use of Random Forest for Prediction

Decision:

Use Random Forest instead of only Linear Regression.

Rationale:

Random Forest handles nonlinear relationships better and provides higher accuracy. It also reduces overfitting compared to single models.

4.2.4. Use of Flask for Web Application**Decision:**

Use Flask instead of complex frameworks.

Rationale:

Flask is lightweight, easy to implement, and suitable for integrating ML models with a web interface.

4.2.5. Real-Time Prediction with Smoothing**Decision:**

Apply smoothing to live camera predictions.

Rationale:

Real-time predictions can fluctuate due to small pose variations. Averaging improves stability and user experience.

4.3 Technical Challenges and Solutions**Challenge 1: Feature Mismatch Between Training and Prediction****Problem:**

The number and names of features did not match between training and prediction.

Solution:

Standardized feature names and ensured consistency across dataset, model, and prediction pipeline

Challenge 2: Incorrect Scaling of Features

Problem:

Extracted features from images were not aligned with dataset values.

Solution:

Introduced scaling factors and normalization to match real-time data with training data.

Challenge 3: Model Bias Toward Single Feature (Hip Width)

Problem:

Model relied heavily on one feature, leading to incorrect predictions.

Solution:

Balanced dataset and adjusted feature relationships to ensure equal contribution.

Challenge 4: Pose Detection Failure

Problem:

Model failed when full body was not visible.

Solution:

Added validation checks for landmark visibility before processing.

Challenge 5: Real-Time Prediction Instability**Problem:**

BMI values fluctuated rapidly in live camera mode.

Solution:

Implemented moving average smoothing for stable output.

4.4 Security Considerations**4.4.1. Safe Handling of User Input**

All uploaded images are processed locally and not used for any external operations.

4.4.2. Prevention of Invalid Data Processing

The system validates input before processing to prevent errors

4.4.3. Secure Model Usage

The trained model is stored locally and not exposed externally

4.4.4. No Execution of User Data

User input is treated only as data and never executed as code.

4.5 Risk Analysis and Mitigation

Risk Table

| Risk ID | Description | Probability | Impact | Risk Level |
|---------|---------------------------|-------------|--------|------------|
| R1 | Poor pose detection | Medium | High | High |
| R2 | Incorrect feature scaling | Medium | High | High |
| R3 | Model overfitting | Low | Medium | Medium |
| R4 | Real-time lag | Medium | Medium | Medium |
| R5 | Incorrect predictions | Medium | High | High |

Risk Mitigation

- Improve pose detection using better validation
- Normalize features properly
- Use multiple models for comparison
- Optimize real-time processing
- Balance dataset

4.6 Risks That Occurred

Risk 1: Feature Mismatch Error

Issue:

Model expected different number of features.

Solution:

Re-trained model with correct feature set.

Risk 2: Underweight Bias

Issue:

All predictions were underweight initially.

Solution:

Improved dataset distribution and feature scaling.

Risk 3: Runtime Errors in Code

Issue:

Errors like indentation and missing variables.

Solution:

Debugging and proper code structuring.

Conclusion

This chapter presented the system design, implementation decisions, challenges, and risk analysis. The system successfully integrates computer vision and machine learning to provide a practical BMI estimation solution.

CHAPTER 5

TESTING

5.1 Testing Plan

5.1.1 Testing Objectives

The testing strategy for the AI-Based BMI Estimation System is designed to ensure that the system is **accurate, reliable, and user-friendly**. The main objectives include:

- Verifying the correctness of BMI prediction using machine learning models
- Ensuring pose detection works accurately for different body positions
- Validating feature extraction logic and consistency with training data
- Evaluating performance of real-time camera-based BMI estimation
- Ensuring smooth interaction between components (MediaPipe, ML model, Flask)
- Identifying and handling errors such as invalid input or incomplete body detection
- Ensuring system stability under continuous usage

5.1.2 Testing Scope

In Scope:

- Image upload and processing
- Live camera BMI estimation
- Pose detection using MediaPipe
- Feature extraction and normalization
- Machine learning model prediction
- BMI classification and result display
- History tracking and storage
- Backend processing and API handling
- Error handling (no body detected, invalid image, etc.)

- User interface functionality

5.1.3 Testing Types and Strategy

Table 5.1: Testing Types Detail

| Type of Test | Will Test Be Performed? | Comments/Explanation |
|---------------------|-------------------------|--|
| Unit Testing | Yes | Individual functions like feature extraction and prediction tested |
| Integration Testing | Yes | Interaction between pose detection, feature extraction, and ML model |
| System Testing | Yes | End-to-end testing of entire BMI system |
| Performance Testing | Yes | Response time for image and live camera prediction |
| Stress Testing | Yes | Continuous camera usage and multiple inputs |
| Security Testing | No | Not required for local deployment |
| Usability Testing | Yes | User interface and experience validation |

Table 5.2: Test Team Details

| Role | Name | Responsibilities |
|------------------|-------------------|---------------------------------|
| Test Lead | Project Developer | Test design and validation |
| Developer/Tester | Project Developer | Unit and integration testing |
| Beta Testers | 2–3 Users | Real-world testing and feedback |

Table 5.3: Test Schedule

| Activity | Start | End | Hours | Comments |
|---------------------|--------|--------|-------|--|
| Unit Testing | Week 2 | Week 4 | 8 | Feature extraction and model functions |
| Integration Testing | Week 4 | Week 6 | 6 | End-to-end system validation |
| Performance Testing | Week 6 | Week 7 | 4 | Real-time camera evaluation |
| User Testing | Week 7 | Week 8 | 6 | UI and usability feedback |

| | | | | |
|--------------------|--------|--------|---|-----------------|
| Regression Testing | Week 8 | Week 9 | 5 | After bug fixes |
|--------------------|--------|--------|---|-----------------|

Table 5.4: Component Testing

| Component | Type of Testing | Technique |
|-----------------------|---------------------|-------------------------------------|
| Pose Detection Module | Unit, Integration | Black Box (input-output validation) |
| Feature Extraction | Unit Testing | White Box (logic testing) |
| BMI Prediction Model | Unit, Performance | Black Box |
| Flask Backend | Integration Testing | API Testing |
| UI Interface | System Testing | Usability Testing |
| History Storage | Integration Testing | Data validation |

5.2 Test Cases

Feature Extraction Module

Equivalence Classes:

- EC1: Valid full-body image

- EC2: Partial body image
- EC3: No human detected
- EC4: Incorrect image format

Table 5.5: Feature Extraction Tests

| Test Case ID | Input | Expected Output | Status |
|--------------|-----------------------|------------------------------|--------|
| FE-001 | Valid full-body image | Features extracted correctly | Pass |
| FE-002 | Partial body image | Error message displayed | Pass |
| FE-003 | No person | “No pose detected” | Pass |
| FE-004 | Invalid image | Error handled gracefully | Pass |

Table 5.6: BMI Prediction Module

| Test Case ID | Input | Expected Output | Status |
|--------------|------------------|----------------------|--------|
| BMI-001 | Valid features | Correct BMI value | Pass |
| BMI-002 | Extreme values | Reasonable BMI range | Pass |
| BMI-003 | Missing features | Error handled | Pass |

5.3 Limitations of the Solution

5.3.1. Dependency on Image Quality

Limitation:

The system requires clear and full-body images for accurate prediction.

Impact:

Low-quality images may lead to incorrect BMI estimation.

Mitigation:

User guidance and validation checks are implemented.

5.3.2. Pose Detection Constraints

Limitation:

MediaPipe may fail if the body is not fully visible.

Impact:

Feature extraction fails, leading to no prediction.

Mitigation:

Visibility checks and error messages are used.

5.3.3. Synthetic Dataset Limitation

Limitation:

Dataset is artificially generated.

Impact:

May not perfectly represent real-world body variations.

Mitigation:

Balanced dataset and realistic feature relationships used.

5.3.4. Model Generalization

Limitation:

Model may not perform equally for all body types.

Impact:

Prediction accuracy may vary.

Mitigation:

Multiple models tested and best selected.

5.3.5. Real-Time Prediction Variability

Limitation:

Camera-based predictions may fluctuate.

Impact:

Unstable BMI output.

Mitigation:

Smoothing techniques applied.

5.3.6. Limited Medical Accuracy

Limitation:

System is not a medical-grade tool.

Impact:

Cannot replace clinical diagnosis.

Mitigation:

Used only for general health estimation.

Conclusion

This chapter describes the testing strategy, test cases, and limitations of the system. The testing ensures that the BMI estimation system performs reliably and meets the required objectives.

CHAPTER 6

OUTPUTS, CONCLUSION AND FUTURE WORK

6.1 OUTPUTS

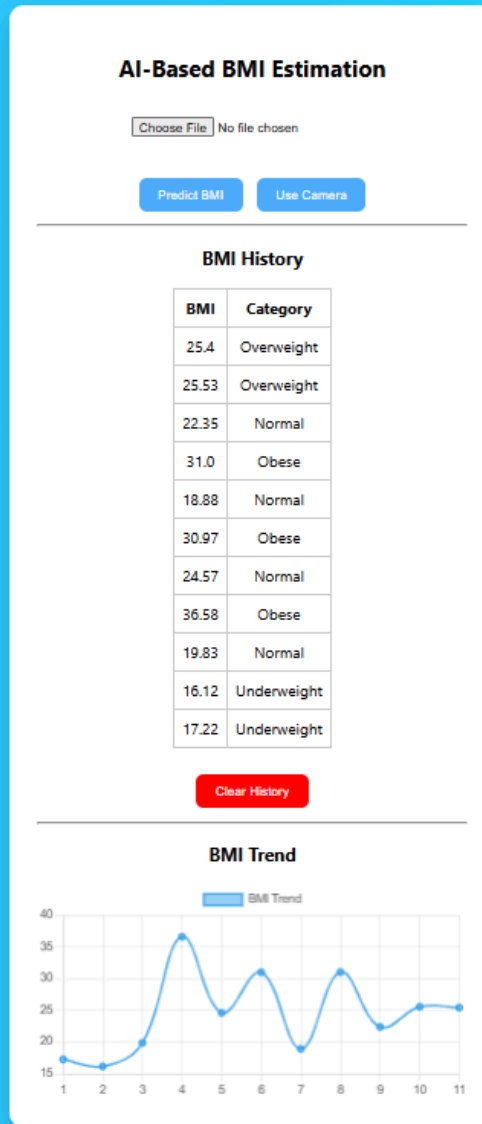


Fig 6.1: Landing Page

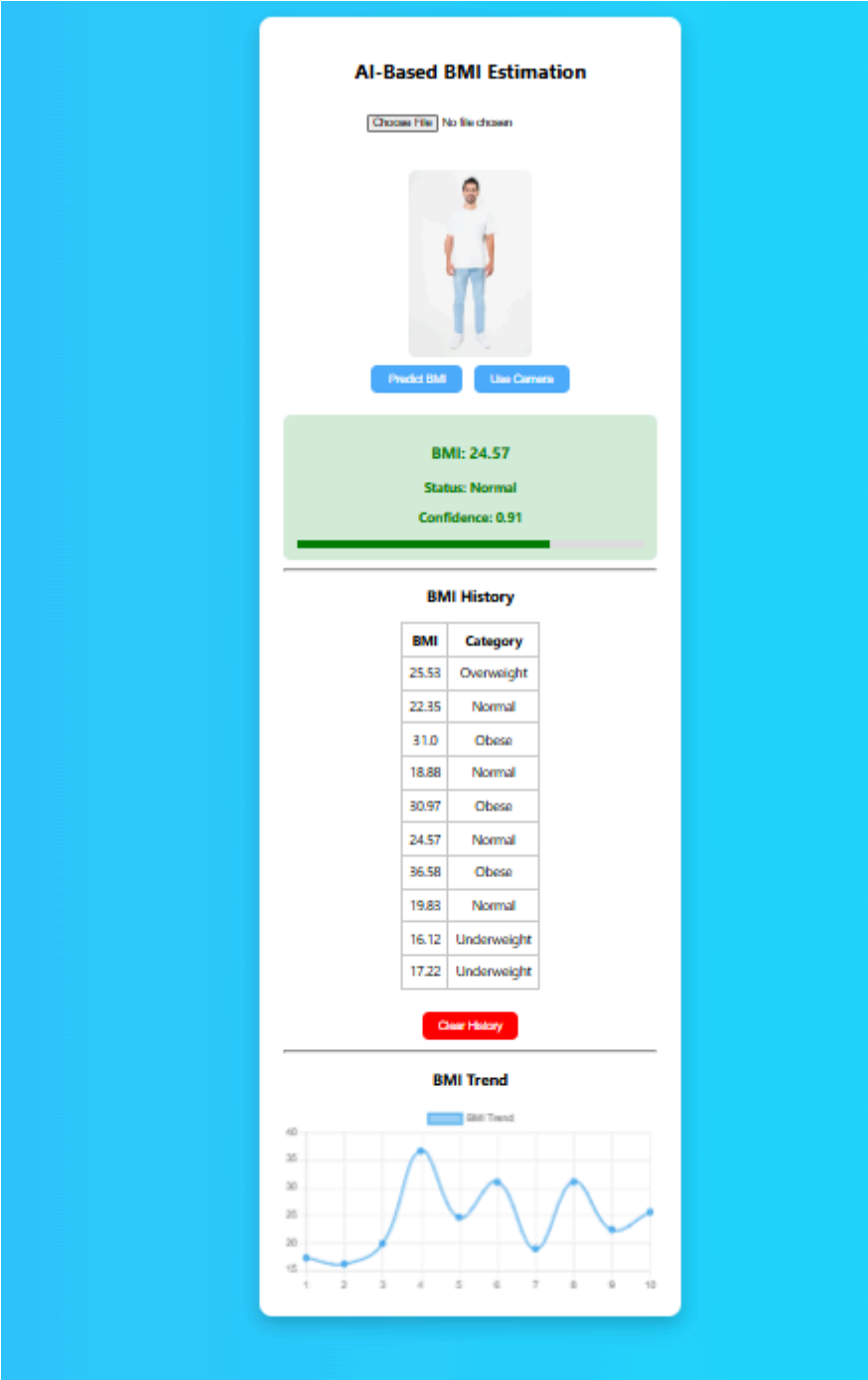


Fig 6.2: Landing screen

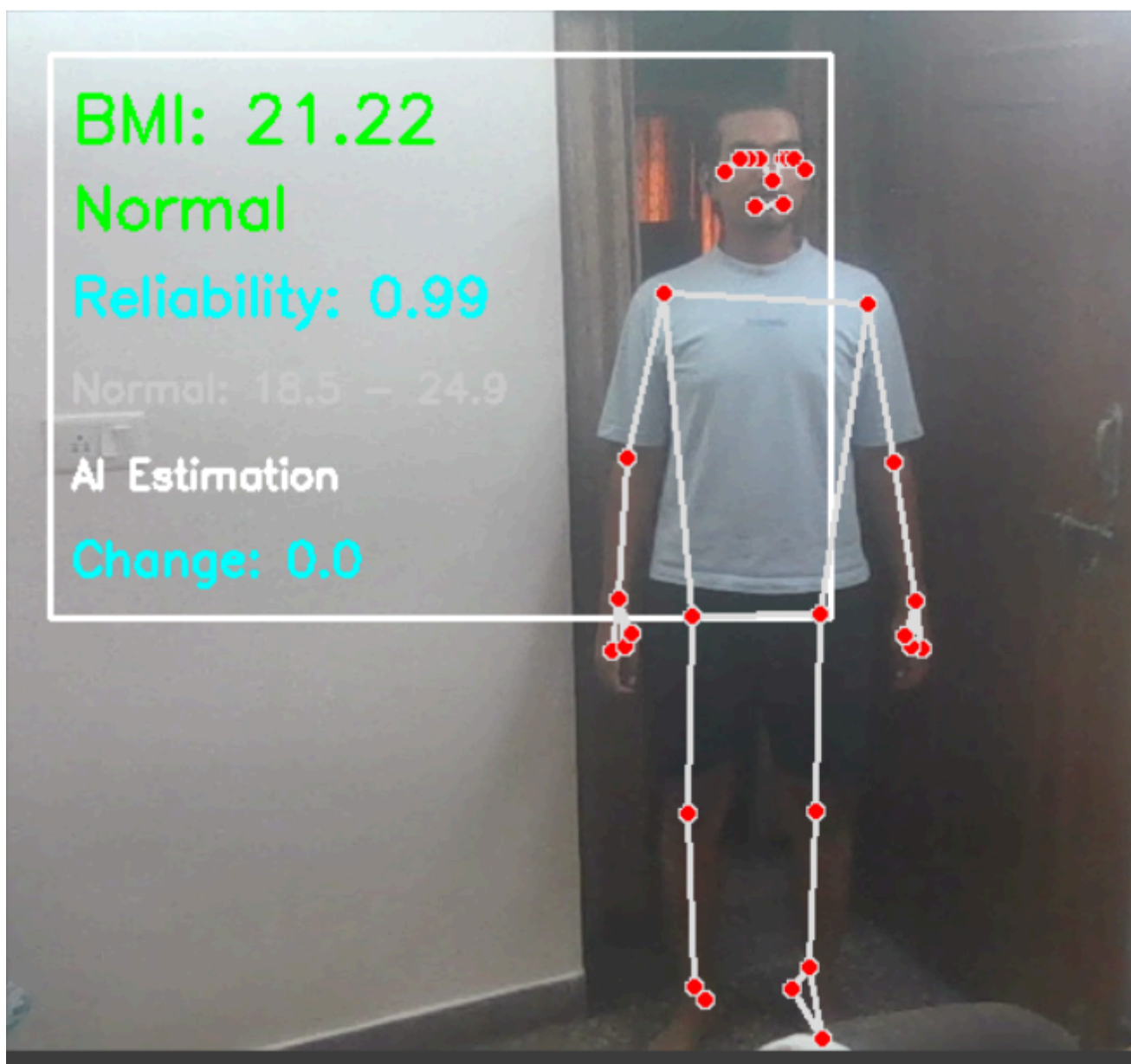


Fig 6.3 Live Camera UI

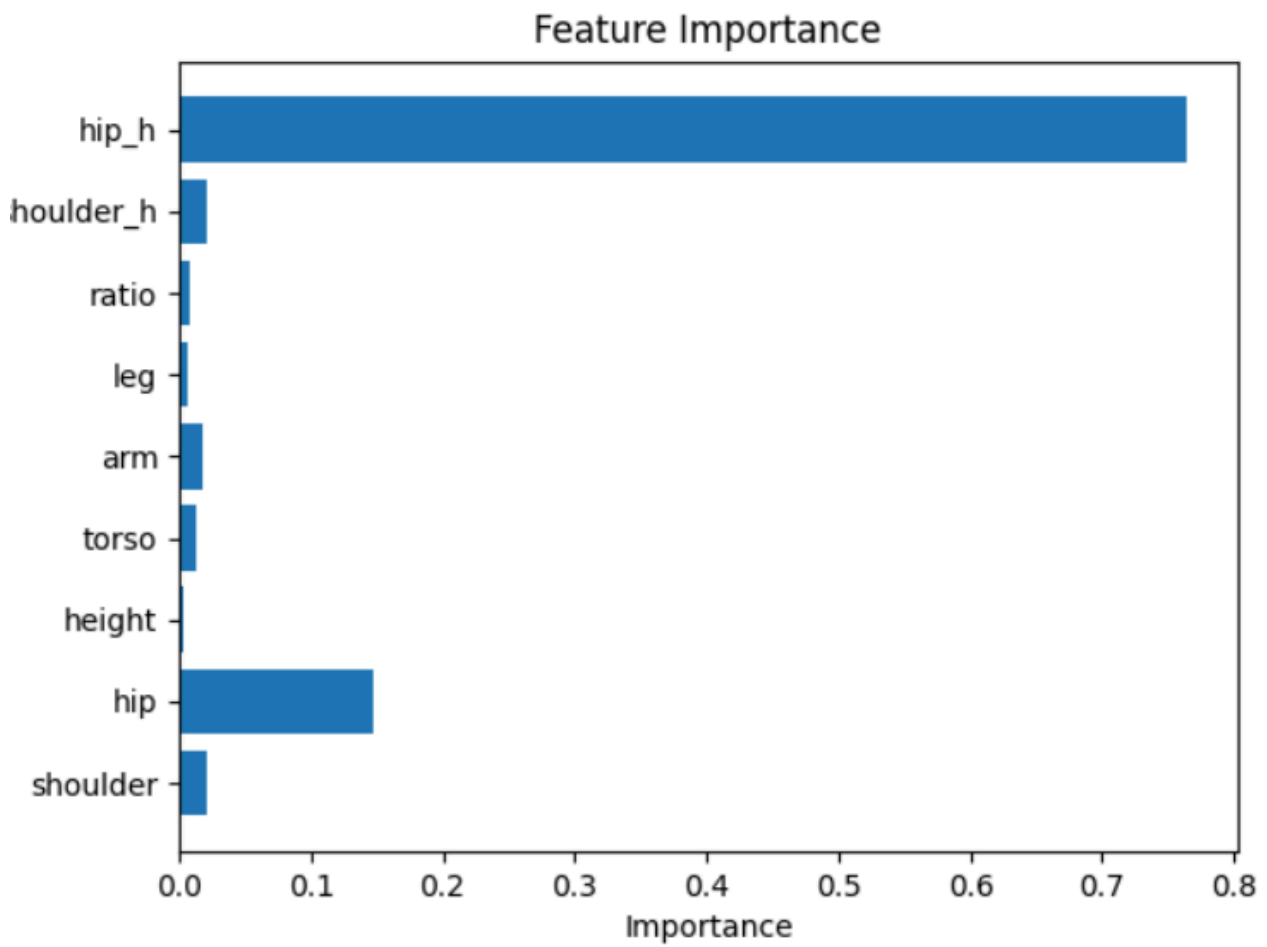


Fig 6.4 Important Feature

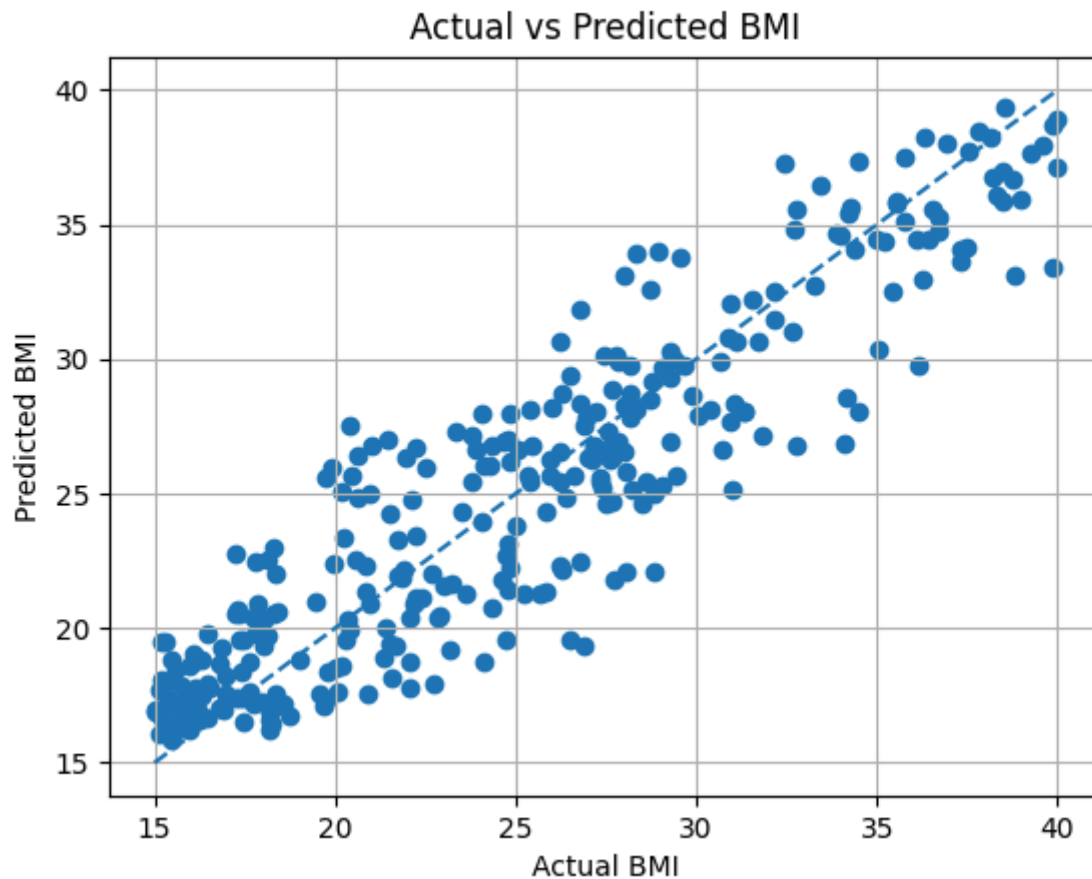


Fig 6.5 evaluation graph

6.2 Conclusion

The AI-Based BMI Estimation system demonstrates how computer vision and machine learning can be combined to create a practical, real-time health assessment tool. Instead of relying on manual measurements such as weight and height, the system uses pose estimation and body proportions extracted from images or live camera input to estimate BMI efficiently.

One of the key strengths of the system is its ability to interpret human body structure using landmark detection. By extracting features such as shoulder width, hip width, torso length, and limb proportions, the model learns meaningful relationships between body shape and BMI. This allows the system to provide quick predictions without requiring physical contact or specialized equipment.

The integration of a machine learning model ensures that predictions improve with better data design. By refining the dataset to reflect realistic human body proportions and fat distribution patterns, the system becomes more accurate and reliable. The use of feature engineering and normalization further enhances prediction stability, especially in real-time scenarios.

The live camera functionality adds significant value by enabling continuous BMI estimation. With smoothing techniques and confidence scoring, the system avoids noisy predictions and provides a more user-friendly experience. Additionally, maintaining a history of predictions allows users to track trends over time.

From a technical perspective, the project highlights important concepts such as data generation, feature alignment, model training, and real-time inference. It also demonstrates how different components—pose estimation (MediaPipe), machine learning (Random Forest), and web interface (Flask)—can be integrated into a complete application.

Overall, the project shows how AI can simplify health monitoring and make it more accessible. It represents a step toward non-invasive, intelligent health tools that can assist users in understanding their physical condition in a fast and convenient manner.

6.3 Future Work

While the current system performs well, several improvements can further enhance accuracy, usability, and real-world applicability.

6.3.1. Improved Dataset Using Real Images

Motivation: Current dataset is synthetic.

Approach:

- Use real annotated body images
- Include different body types, genders, and age groups
- Combine synthetic + real data

Benefits:

Higher accuracy
Better generalization
More realistic predictions

6.3.2. Gender and Body-Type Awareness

Motivation: Body fat distribution differs across individuals

Approach:

- Add gender classification
- Model apple vs pear body shapes
- Adjust feature importance dynamically

Benefits:

- More personalized predictions
- Better medical relevance

6.3.3. Advanced Deep Learning Model

Motivation: Improve prediction quality

Approach:

- Replace RandomForest with CNN-based model
- Use full image instead of only extracted features

Benefits:

- Capture visual patterns
- Higher prediction accuracy

6.3.4. Real-Time Feedback System

Motivation: Improve user interaction

Approach:

- Show posture correction hints
- Guide user to stand properly
- Detect incomplete body visibility

Benefits:

- Better input quality
- More consistent predictions

6.3.5. Mobile Application Integration

Motivation: Increase accessibility

Approach:

- Convert system into Android/iOS app
- Use phone camera for real-time BMI

Benefits:

- Portable solution
- Wider user reach

This project demonstrates how artificial intelligence can move beyond traditional data input methods and provide intelligent, real-time health insights using only visual information, making health monitoring more accessible, scalable, and user-friendly.

REFERENCES

- [1] Google AI, “**MediaPipe Pose: Real-Time Human Pose Tracking**,” 2023. [Online]. Available: [\[https://developers.google.com/mediapipe\]](https://developers.google.com/mediapipe)(<https://developers.google.com/mediapipe>)
- [2] Z. Cao, T. Simon, S. E. Wei, and Y. Sheikh, “**OpenPose: Realtime Multi-Person 2D Pose Estimation**,” IEEE Trans. Pattern Anal. Mach. Intell., 2021.
- [3] OpenCV, “**Open Source Computer Vision Library**,” 2024. [Online]. Available: [\[https://opencv.org\]](https://opencv.org)(<https://opencv.org>)
- [4] TensorFlow, “**TensorFlow Lite for Mobile and Edge Devices**,” 2024. [Online]. Available: [\[https://www.tensorflow.org/lite\]](https://www.tensorflow.org/lite)(<https://www.tensorflow.org/lite>)

Machine Learning

- [5] L. Breiman, “**Random Forests**,” Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.
- [6] F. Pedregosa et al., “**Scikit-learn: Machine Learning in Python**,” J. Mach. Learn. Res., vol. 12, pp. 2825–2830, 2011.
- [7] C. M. Bishop, **Pattern Recognition and Machine Learning**, Springer, 2006.
- [8] I. Goodfellow, Y. Bengio, and A. Courville, **Deep Learning**, MIT Press, 2016.

BMI & Health

- [9] World Health Organization, “**Body Mass Index (BMI) Classification**,” 2023. [Online]. Available: [\[https://www.who.int\]](https://www.who.int)(<https://www.who.int>)

[10] Centers for Disease Control and Prevention, “**About Adult BMI,**” 2024. [Online]. Available: [\[https://www.cdc.gov\]](https://www.cdc.gov)(<https://www.cdc.gov>)

[11] NCD Risk Factor Collaboration, “**Worldwide Trends in Body-Mass Index,**” The Lancet, vol. 400, 2022.

[12] Harvard T.H. Chan School of Public Health, “**BMI and Health Risks,**” 2023.

AI in Healthcare

[13] IEEE, “**Artificial Intelligence in Healthcare,**” IEEE Access, 2022.

[14] Nature, “**Machine Learning for Medical Diagnosis,**” Nature Medicine, 2021.

[15] Springer, **Artificial Intelligence in Healthcare Systems,** Springer, 2023.

Computer Vision in Body Analysis

[16] Computer Vision and Image Understanding, “**Human Body Shape Estimation Using Vision-Based Methods,**” 2022.

[17] IEEE Transactions on Biomedical Engineering, “**Vision-Based Anthropometric Measurement Systems,**” 2021.

Tools & Development

[18] Flask, “**Flask Web Framework Documentation,**” 2024. [Online]. Available: [\[https://flask.palletsprojects.com\]](https://flask.palletsprojects.com)(<https://flask.palletsprojects.com>)

[19] Python Software Foundation, “**Python Language Documentation**,” 2024. [Online]. Available: [\[https://www.python.org\]](https://www.python.org)(<https://www.python.org>)

[20] NumPy Developers, “**NumPy User Guide**,” 2024. [Online]. Available: [\[https://numpy.org\]](https://numpy.org)(<https://numpy.org>)